→ Design Hashset

① Using List :

so what we can do is, have a huge as list

Now each position in the list will belong to a key

For eg:- key = 4

0	1	2	3	4	5	6	7
0	0	0	0	4	0	0	0

so at idx 4, we add key 4

similarly key = 6

0	1	2	3	4	5	6	7
0	0	0	0	4	0	6	0

* So insert can be done like list[key] = key

* remove : list[key] = 0 → Just make the Value at that place as 0

* Contains : if list[key] != 0 : return True

Problem

→ Range will Create problem

- It is easy to create list maybe upto 10^7
- Anything $> 10^7$: It becomes really difficult to perform operations

Solution :

Since Range is a problem. We can minimize the size of List with a hash function.

There are many hash functions available, easiest One among is

⇒ % Mod function

so If we have key = 20, Len of list = 10, the we can add 20 % 10 = 0

0	1	2	3	4	5	6	7	8	9

so at Idx 0 we can add 20

⇒ Problem : Collision : what if we want to add multiple Value at same idx, For Eg : 20, 30, 40, 50 Everything will go to idx 0.

So to prevent collision, there are various way, One such way is Linked List

Now at Idx = 0, we can have a chain of number

so
- Insert = Insert as Last Node in LL
- remove = remove node if present
- contain = check if node is present